

DIGITAL FORENSIC EXAMINATION REPORT

UNMANNED AERIAL VEHICLE (UAV) FLIGHT DATA & TELEMETRY EXTRACTION

1. INVESTIGATIVE CASE ADMINISTRATION

Case Reference:	CASE-2026-MEITY-003	Evidence Item ID:	4c0030a3-fcb9-4252-968a-11256166c543.ulg
Examining Agency:	Cyber Forensic Investigation Laboratory (CFSL / State Police)	Lead Examiner:	Inspector Cyber Division1
Examination Date:	2026-09-05 09:01:54 UTC	Forensic Engine:	Pushpak Indigenous Drone Forensics Toolkit (v1.0.0-phase1 (Build 2026.09))

2. EVIDENCE INTEGRITY & CHAIN OF CUSTODY

Source Evidence File:	C:\Users\pawan\Downloads\4c0030a3-fcb9-4252-968a-11256166c543.ulg
File Size:	25,645,967 bytes
SHA-256 Digest:	b164b83b5c542ee65743e011cfc04aef5e1c557e8dd228083bea0d796c982b75
BLAKE3 Digest:	4548bd101b421a173789ebdcf912608cb43488b21f5beb3661137cd8fb85ac30
Chain Verification:	VERIFIED INTACT (100% Cryptographic Match)

3. FLIGHT MISSION ENVELOPE & TELEMETRY PROFILE

Platform / Architecture:	PX4	Aircraft Model:	PX4_FMU_V6X
Hardware Serial No:	N/A	Total Telemetry Records:	5,904 events
Takeoff Time (UTC):	2026-08-22 06:04:29 UTC	Termination Time (UTC):	2026-08-22 06:13:22 UTC
Total Flight Duration:	532.5 seconds (8.9 mins)	Total GPS Waypoints:	4,242 points
Max Altitude (AGL):	494.1 meters (1621.2 ft)	Peak Ground Velocity:	7.5 m/s (26.9 km/h)

4. FORENSIC ANOMALY FINDINGS & THREAT ASSESSMENT

Anomaly Type	Severity	Timestamp (UTC)	Location / Alt	Technical Finding Narrative
GPS SIGNAL DEGRADATION	MEDIUM	06:04:29 UTC	N/A	GPS lock compromised: only 0 satellites visible
GPS SIGNAL DEGRADATION	MEDIUM	06:04:29 UTC	N/A	GPS lock compromised: only 0 satellites visible
GPS SPOOFING OR TELEPORTATION	HIGH	06:04:33 UTC	30.4047, 104.1134 Alt: 476.0m	Impossible kinematic displacement: calculated velocity 3087081.3 m/s exceeds airframe limit (45.0 m/s) over 3.68s
GPS SPOOFING OR TELEPORTATION	HIGH	06:04:33 UTC	N/A	Impossible kinematic displacement: calculated velocity 118685426.9 m/s exceeds airframe limit (45.0 m/s) over 0.10s

GPS SIGNAL DEGRADATION	MEDIUM	06:04:33 UTC	N/A	Severe GPS geometry degradation: HDOP reached 99.90
GPS SPOOFING OR TELEPORTATION	HIGH	06:04:33 UTC	30.4047, 104.1134 Alt: 474.5m	Impossible kinematic displacement: calculated velocity 108887242.0 m/s exceeds airframe limit (45.0 m/s) over 0.10s
GPS SIGNAL DEGRADATION	MEDIUM	06:04:34 UTC	N/A	Severe GPS geometry degradation: HDOP reached 99.90
GPS SPOOFING OR TELEPORTATION	HIGH	06:04:34 UTC	30.4047, 104.1134 Alt: 470.0m	Impossible kinematic displacement: calculated velocity 71026011.4 m/s exceeds airframe limit (45.0 m/s) over 0.16s
GPS SIGNAL DEGRADATION	MEDIUM	06:04:35 UTC	N/A	Severe GPS geometry degradation: HDOP reached 99.90
GPS SPOOFING OR TELEPORTATION	HIGH	06:04:35 UTC	30.4047, 104.1134 Alt: 466.0m	Impossible kinematic displacement: calculated velocity 71441513.3 m/s exceeds airframe limit (45.0 m/s) over 0.16s
GPS SPOOFING OR TELEPORTATION	HIGH	06:04:36 UTC	N/A	Impossible kinematic displacement: calculated velocity 81654488.5 m/s exceeds airframe limit (45.0 m/s) over 0.14s
GPS SIGNAL DEGRADATION	MEDIUM	06:04:36 UTC	N/A	Severe GPS geometry degradation: HDOP reached 99.90

5. FORENSIC CONFIDENCE & UNCERTAINTY ASSESSMENT

Overall Forensic Confidence Rating: HIGH (HIGH COURT ADMISSIBLE) - 99.0%
In accordance with ISO/IEC 27037:2012 guidelines, this assessment explicitly defines the mathematical certainty and evidentiary boundaries of the reconstructed flight profile.

Forensic Dimension	Confidence Level	Evidentiary Basis & Limitations
Bit-Stream Integrity	ABSOLUTE (100%)	Dual SHA-256 + BLAKE3 hashes match original media bit-stream.
Spatial Trajectory	HIGH (99.0%)	Avg Satellites: 25.7 Avg HDOP: 0.85 Multipath risk low.
Temporal Synchronization	HIGH (99.0%)	Correlated with GPS atomic UTC reference clock.
Evidence Limitations	DOCUMENTED	Barometric altitude uncorrected for local QNH pressure.

6. CHRONOLOGICAL MILESTONE TIMELINE

Timestamp (UTC)	Event Type	Mode / State	Forensic Evidence Context
2026-08-22 06:04:29	GPS FIX	N/A	Telemetry fix
2026-08-22 06:04:31	MODE CHANGE	POSCTL	Telemetry fix
2026-08-22 06:04:31	ARM DISARM	N/A	Telemetry fix
2026-08-22 06:04:33	ARM DISARM	N/A	[health_and_arming_checks] Preflight Fail: GNSS heading invalid
2026-08-22 06:04:33	ARM DISARM	N/A	[health_and_arming_checks] Preflight Fail: No connection to the ground control station
2026-08-22 06:04:35	ARM DISARM	N/A	[health_and_arming_checks] Preflight: GPS Horizontal Pos Drift too high

2026-08-22 06:04:35	ARM DISARM	N/A	[health_and_arming_checks] Preflight Fail: GNSS heading invalid
2026-08-22 06:04:35	ARM DISARM	N/A	[health_and_arming_checks] Preflight Fail: No connection to the ground control station
2026-08-22 06:04:41	RTH TRIGGER	N/A	[navigator] Flight time and intelligent return-to-home update anomaly,land immediately for inspection!!!.
2026-08-22 06:04:42	ARM DISARM	N/A	[health_and_arming_checks] Preflight: GPS Horizontal Pos Drift too high
2026-08-22 06:04:42	ARM DISARM	N/A	[health_and_arming_checks] Preflight Fail: GNSS heading invalid
2026-08-22 06:04:44	ARM DISARM	N/A	[health_and_arming_checks] Preflight: GPS Vertical Pos Drift too high
2026-08-22 06:04:44	ARM DISARM	N/A	[health_and_arming_checks] Preflight Fail: GNSS heading invalid
2026-08-22 06:04:46	ARM DISARM	N/A	[health_and_arming_checks] Preflight: GPS Horizontal Pos Drift too high
2026-08-22 06:04:46	ARM DISARM	N/A	[health_and_arming_checks] Preflight Fail: GNSS heading invalid

7. CERTIFICATE OF AUTHENTICITY (LEGAL ADMISSIBILITY)

I hereby certify under Section 63 of the Bharatiya Sakshya Adhiniyam, 2023 (corresponding to Section 65B of the Indian Evidence Act, 1872) and ISO/IEC 27037 standards that the electronic records, telemetry chronologies, and mathematical calculations contained within this report were generated by lawful digital forensic acquisition. The computing systems and software tools used to parse this evidence operated properly throughout the examination with zero alteration to original electronic memory. The cryptographic digests (SHA-256 and BLAKE3) confirm absolute bit-stream integrity of the exhibited evidence.

Examiner Signature: _____

Date: 2026-09-05

Name: Inspector Cyber Division1

Title: Lead Digital Forensics Examiner

Agency: Cyber Forensic Investigation Laboratory (CFSL / State Police)

Official Seal: [AFFIX LABORATORY SEAL]